Trinetra Project

Software Requirement Specification Document

Trinetra



	Trinetra Wireless
	Standard API

Standard API Document

API

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1. Introduction

The purpose of this document is to brief the details about the API to pull the data from the Trinetra server.

2. Product Overview

Trinetra uses industry leading web technologies and cutting-edge hardware design to bring you the highest quality of **Fleet Management, Mobility & IoT Solutions**.

Trinetra Wireless is an innovative technology company with proven domain expertise, an advanced hardware design center in Canada, an excellent embedded engineering and software development center in India. With sales offices in 8+ global locations, we also work in partnership with channel partners and system integrators worldwide to provide quality solutions to our clients.

More information on **Trinetra** is available at www.trinetrawireless.com

Trinetra offers unique advantages such as:

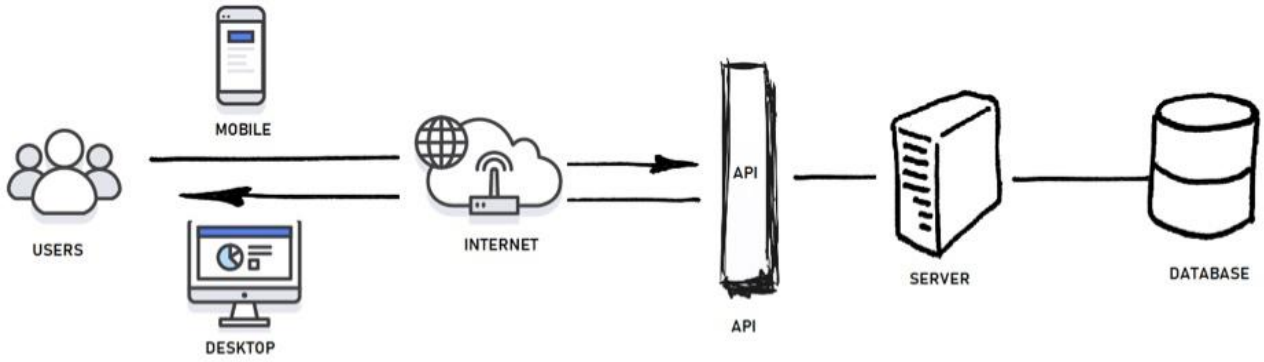
- An integrated offering of GPS hardware and software
- A fully hosted solution requiring no software installation
- Real Time Location and Status Information
- Sensor Integration
- Over-the-Air-Firmware upgrade
- Proprietary protocol enabling ease of customization

3. Important Note:

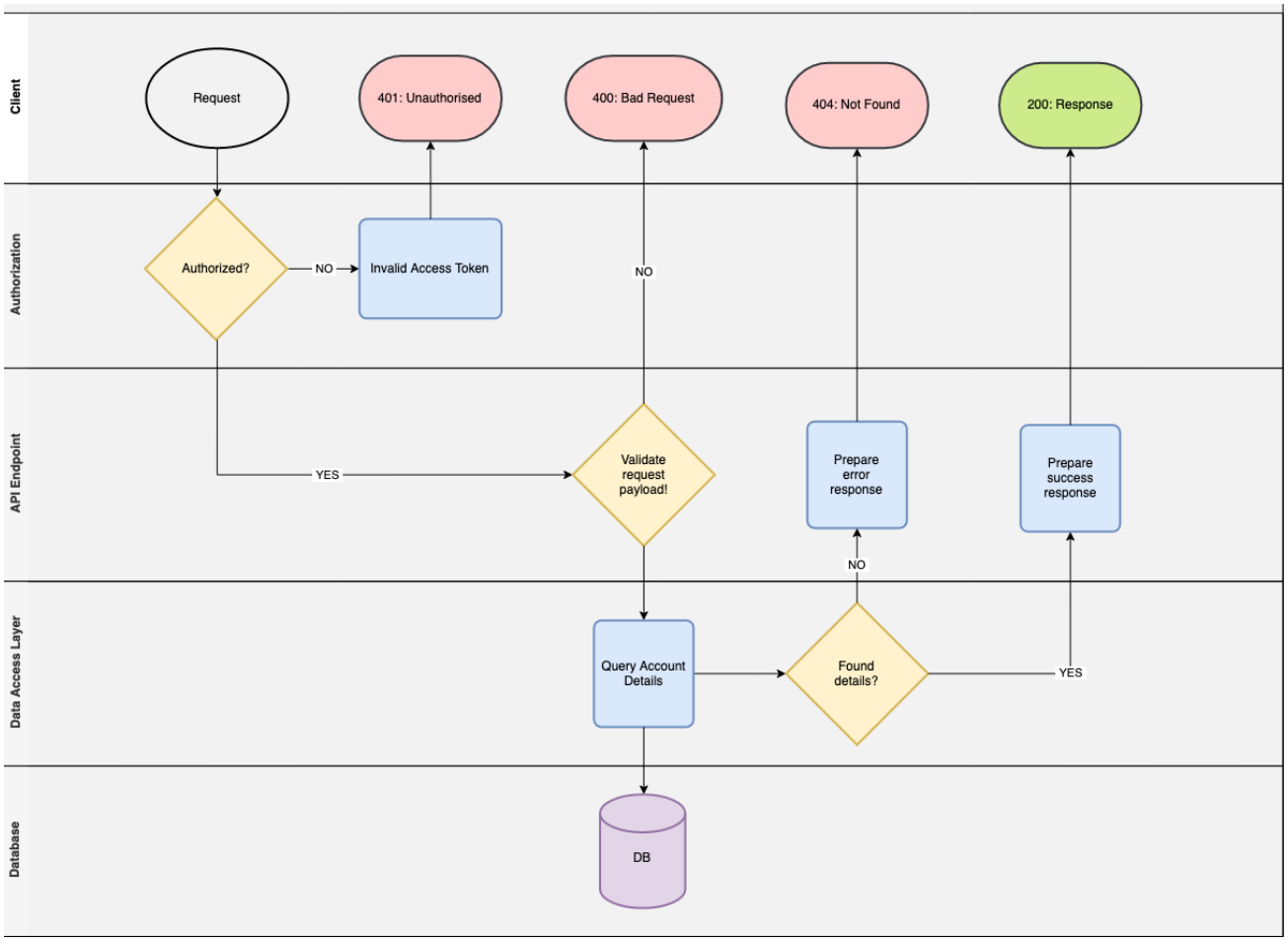
- **API Endpoint Standardization**
    - For APIs, we provide standard endpoints and the data update request will be typically at configured interval. You can access these APIs to collect data for integration purposes. If you are referring to our Trinetra device, the reporting interval (e.g., every 5 minutes) determines how often data is sent to our application
- **Optimizing API Usage:**
    - Typically, the API hits are at an interval of every 5 minutes once and the data will be updated based on the reporting interval of the device. However, there are no specific limits but there is a nominal charge based on number of calls volume.
- **Service Level Agreements (SLAs):**
    - High Priority / Crash will be within 1 business day

- Medium Priority will be 2 - 3 business days
- Low Priority will be 4-5 business days week

4. Architecture



5. Flow chart



6. Login

This API is used to get the User details with token of access the Trinetra API system. Once the API is hit then new token will be generated, after all request the token should be must. This API includes the filter parameters and Authentication token for the client has to pass as part of the header. The token expiration duration is set to **7 days** from the time of issuance.

API End Point:

https://www.trinetrafms.com/TriWebapi/api/Login/Userlogin

Header:

Content-Type: application/json

Parameter	Type	Mandatory	Notes
[UserName]	String	Yes	User name of Trinetra portal.
[Password]	String	Yes	Password of Trinetra portal.
[OrganizationCode]	String	Yes	Registered organization code.

Sample:

[{"key":"Content-Type","value":" application/json","description":""}]

Request Payload:

```
{
  "userName": [UserName],
  "password": [Password],
  "orgcode": [OrgCode]
}
```

The response to the above request provides the following elements

```
{
    "userId": 17,
    "userName": "karthick",
    "displayName": "karthick(R)",
    "organizationDisplayName": null,
    "organizationId": 1,
    "email": "karthikeyan.r@angleritech.com",
    "expiryFlag": 0,
    "expiryDays": 0,
     "token":
"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJodHRwOi8vc2NoZW1hcy54bWxzb2FwLm9yZy93cy
8yMDA1LzA1L2lkZW50aXR5L2NsYWltcy9uYW1laWRlbnRpZmllciI6IjE3IiwiaHR0cDovL3NjaGVtY
XMueG1sc29hcC5vcmcvd3MvMjAwNS8wNS9pZGVudGl0eS9jbGFpbXMvbmFtZSI6ImthcnRoaW
NrIiwiaHR0cDovL3NjaGVtYXMueG1sc29hcC5vcmcvd3MvMjAwNS8wNS9pZGVudGl0eS9jbGFpb
XMvZW1haWxhZGRyZXNzIjoia2FydGhpa2V5YW4uckBhbmdsZXJpdGVjaC5jb20iLCJodHRwOi8v
c2NoZW1hcy54bWxzb2FwLm9yZy93cy8yMDA1LzA1L2lkZW50aXR5L2NsYWltcy9zaWQiOiIxNyIs
Imp0aSI6IjZkMjc3Y2UxLTM5NmItNDllYy05MDE2LTcxYWQ1ZWI5NGRjMCIsImV4cCI6MTc3NTkw
NzE2OSwiaXNzIjoiVGVzdC5jb20iLCJhdWQiOiJUZXN0LmNvbSJ9.ck0ugWhM-
4qyd7ZJ6eLnLxN_9kL7gx9ldPHwGQDkA7E",
     "responseStatus": true,
     "responseMessage": "Success"
}
```


7. History

This API is used to get the history details of fleets from Trinetra system. The History API for vehicle tracking is a component of a vehicle tracking system that allows developers to access and manage historical location data of vehicles. This API enables tracking system administrators, fleet managers, or end-users to retrieve detailed information about a vehicle's past locations and activities. This API includes the filter parameters and Authentication token for the client has to pass as part of the header.

API End Point:

https://www.trinetrafms.com/TriWebapi/api/History/getVehicleHistory

Authorization:

Token: <token>

Header:

X-ClientId: [**Trinetra Will Provide**]
Content-Type: application/json

Parameter	Type	Mandatory	Notes
[Vehiclename]	String	Yes	Vehicle name as configured in Trinetra portal.
StartDate	Date Time	Yes	Date filter From date. Date Format: yyyy-MMM-dd hh:mm:ss AM/PM
EndDate	String	Yes	Date filter To date. Date Format: yyyy-MMM-dd hh:mm:ss AM/PM

Sample:

[{"key":" X-ClientId ",
"value":"**[Trinetra Will Provide]",**
"description":"",
"type":"text",
"enabled":true},
{"key":"Content-Type","value":" application/json","description":""}]

Request Payload:

```
{
        "Vehiclename":"[VehicleName]",
        "StartDate":"2019-Dec-01 12:00:00 AM",
        "EndDate":"2019-Dec-03 11:59:00 PM"
}
```

The response to the above request provides the following elements


```
{
   "objHistroy": [
      {
         "vehicleId": "[VehicleID]",
         "vehicleDescription": "[VehicleName]",
         "origindatetime": "09/13/2024 18:27:36",
         "latitude": "13.163817",
         "longitude": "80.226837",
         "streetAddress": "[Address]",
         "validGPS": "0",
         "distance": null,
         "messageDetails": "Location Update",
         "odoMeter": "52459",
         "boxID": "[DeviceID]",
         "units": "KMPH",
         "speed": "-",
         "message": "Location Update",
         "responseCode": null
      }
   ],
   "statusMessage": "Success",
   "statusCode": 1
}
```


8. Monitor

This API is used to get the monitor details of their fleets from Trinetra system. The Monitor API for vehicle tracking is a real-time tracking component of a vehicle tracking system that provides developers with access to the current status and location of vehicles. This API allows users to monitor the live whereabouts and operational status of individual vehicles or entire fleets. This API includes the filter parameters and Authentication token for the client has to pass as part of the header.

API End Point:

https://www.trinetrafms.com/TriWebapi/api/Monitor/Monitor

Authorization:

Token: <token>

Header:

X-ClientId: **[Trinetra Will Provide]**
Content-Type: application/json

Parameter	Type	Mandatory	Notes
Vehiclename	String	No	Vehicle name as configured in Trinetra portal.
FleetName	String	Yes	Fleet name as configured in Trinetra portal.

Sample:

[{"key":" X-ClientId",
"value":"[Trinetra Will Provide]",
"description":"",
"type":"text",
"enabled":true},
{"key":"Content-Type","value":" application/json","description":""}]

Request Payload:

```
{
    "Vehiclename": "",
    "FleetName": "All"
}
```

The response to the above request provides the following elements


```
{
  "objMonitor": [
    {
      "deviceID": "350612070177637",
      "vehicleDescription": "DD07F1D0E5",
      "licensePlate": "[LicenseNo]",
      "address": "Ariyalur Sub-District 621713, India",
      "dateTime": "20 Sep 2024  2:29:06 PM",
      "lastLatitude": "11.184875",
      "lastLongitude": "79.101337",
      "lastSpeed": "0",
      "todayKM": "3",
      "odometerValues": "0335087",
      "geozonename": null,
      "geozoneStatus": "OUT",
```

```
                "geoZoneINOUTDatetime": null,
                "vendorName": "-",
                "status": "Parked",
                "temperature1": "-",
                "temperature2": "-",
                "temperature3": "-",
                "temperature4": "-",
                "humidity1": "-",
                "humidity2": "-",
                "humidity3": "-",
                "humidity4": "-"
            }],
            "statusMessage": "Success",
            "statusCode": 1
        }
```


9. Do's and Don'ts'

Do's:

- **Read the Documentation:** Always start by thoroughly reading the documentation provided by the Trinetra. Understand the endpoints, request/response formats, authentication methods, rate limits details.
- **Use Secure Connections:** Ensure that API requests are made over HTTPS to encrypt data transmission and protect sensitive information from interception.
- **Handle Errors Gracefully:** Implement error handling mechanisms to gracefully handle API errors and exceptions. Provide informative error messages to users and log errors for troubleshooting.
- **Authenticate Requests:** Trinetra API requires authentication, ensure that you follow the authentication process correctly. Use API keys methods as specified in the documentation.
- **Respect Rate Limits:** Adhere to rate limits or usage quotas specified by the Trinetra. Throttling requests helps prevent server overload and ensures fair usage of resources.
- **Use Caching:** Implement caching mechanisms to store frequently requested data locally and reduce the number of API calls. Respect cache headers provided by the Trinetra to control caching behavior.
- **Validate Input Data:** Validate all input data before sending requests to the API to prevent injection attacks, data corruption, or unexpected behavior.

Don'ts:

- **Don't Hardcode Credentials:** Avoid hardcoding sensitive information such as API keys or access tokens directly in your code. Instead, use environment variables or secure storage mechanisms.
- **Don't Overload the API**: Avoid making excessive API calls or repeatedly polling for updates if they are not necessary. Respect rate limits and use caching to minimize the load on the API server.
- **Don't Ignore Error Responses:** Always handle error responses returned by the API. Ignoring errors can lead to unexpected behavior or data loss in your application.
- **Don't Expose API Keys:** Take precautions to prevent API keys from being exposed in client-side code, logs, or network traffic. Use server-side code to make authenticated requests whenever possible.
- **Don't Make Unnecessary Requests**: Minimize the number of API requests by batching or consolidating requests whenever possible. Fetch only the data you need to reduce bandwidth usage and improve performance.
- **Don't Abuse APIs:** Avoid abusing or misusing APIs for purposes other than their intended use. Respect the terms of service and usage policies outlined by the Trinetra.

10. Benefits and Conclusion

Benefits

Real-time Tracking: With the Vehicle Tracking API, you can monitor the location and status of vehicles in real-time, enabling efficient fleet management and logistics operations.

Improved Efficiency: By accessing vehicle data through the API, businesses can optimize routes, reduce fuel consumption, and enhance overall operational efficiency.

Enhanced Safety: The API enables the implementation of safety features such as landmark, geozone, speed monitoring, and driver behavior analysis, contributing to safer driving practices and accident prevention.

Cost Savings: Utilizing the Vehicle Tracking API can lead to cost savings through better resource utilization, reduced vehicle downtime, and improved asset management.

Custom Integration: The API allows seamless integration with existing systems and applications, enabling businesses to incorporate vehicle tracking functionality into their workflows and processes.

Data-driven Decision Making: Access to vehicle tracking data through the API empowers businesses to make informed decisions based on real-time insights, leading to improved planning and resource allocation.

Enhanced Customer Experience: By providing accurate tracking information to customers, businesses can enhance the overall customer experience, leading to increased satisfaction and loyalty.

Conclusion:

In conclusion, the Vehicle Tracking API offers numerous benefits for businesses seeking to optimize their fleet operations and improve efficiency. By providing real-time tracking, enhanced safety features, cost savings, and custom integration capabilities, the API enables businesses to streamline their operations, reduce costs, and make data-driven decisions. With its ability to enhance safety, efficiency, and customer experience, the Vehicle Tracking API is a valuable tool for businesses looking to stay competitive in today's fast-paced transportation industry.

With Best Regards,

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